DATA SCIENCE 2VG3 Homework 6 2025/2026
Due: End of Day, Friday, 19th March 2026, via Avenue

Collisions with Sequential Impulses

Hand the final code in on avenue that contains all test cases that can be run by changing the scenario variable. For the written answers and discussion, hand-in a separate pdf file with your answers.

Exercise 1: **Complete your rigidbody2d_seq.py code as discussed and worked on in class to include the sequential impulse solver.** (see VG3_MultipleContacts.pdf lecture notes). You should have begun with rigidbody2d_seq0.py from Avenue or similar code.

Make sure to include that your solution includes a velocity bias correction:

```
vBias = dclose*0.3/dt
vNormal.append( -self.eCoeffRestitution*uNormal + vBias )
```

Exercises 2: Set up and run the following tests:

(a) scenario=0: Block on an inclined plane sin(angle) = 0.1. Restitution e=0.05, friction coefficients 0.5, gravity on: It should stop!
(b) scenario=1: Block hitting a flat plane with gravity off. Make sure the velocity is straight down (0,0,-1). Restitution e=0.0. It should stop!

Describe what you see in detail. Explore how many iterations are needed to get what you consider to be good solutions. What goes wrong for small numbers of iterations?

Exercise 3: Block bouncing off a plane with gravity off. Restitution e=1.0

(a) scenario=2: Block hitting an inclined plane sin(angle) = 0.1. Make sure the velocity is (-sin(angle), 0, -cos(angle)) initially. It should bounce off.
(b) scenario=3: Block hitting a flat plane (angle=0). Make sure the velocity is straight down (0,0,-1) initially. It should bounce off.

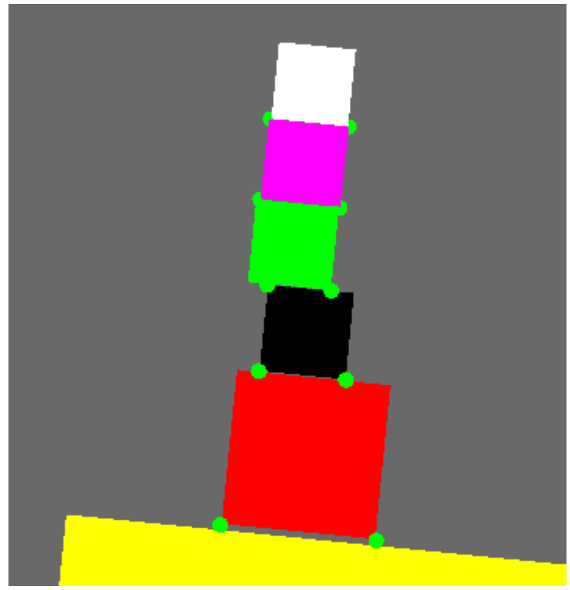
Describe what you see in detail. If everything is correct, it should bounce off exactly the way it came in. Use a decent number of iterations (as determined previously). Measure the total energy before and after the collision.

Exercise 4: Block bouncing repeatedly off a flat plane with gravity on. Restitution e=1.0. No friction.

(a) scenario=4: For this case use the block hitting a flat plane (Make sure the velocity is straight down (0,0,-1). It should bounce forever!

Describe what happens and look at conservation of energy (including gravity). What is the impact of the velocity bias?

Exercise 5: Stacking 5 boxes on a plane.



(a) scenario=5: There is code provide in the lecture slides to make a slightly off centre stack.

Implement a stack according to the instructions in the lectures (should look like the image on the right when it is stable. Box colours don't matter).   The stack should not just be neatly centred but include some off centred boxes as presented in lectures.   Test it out for gravity on, e=0.05 and friction on (coefficients=0.2) for an inclined plane (sin(angle) = 0.1).

You should be able to get it to stay in a stack. Experiment with parameter choices (e.g. bias and iterations) but do not change the physical set up (angle, restitution, friction or gravity).   A better outcome would include less squeezing, less jitter and minimal overlap at any point but particularly at the end.   Report your optimal parameter choices and a *screenshot* of your best stable stack!

Velocity bias if on all the time by default. Box2D's author suggests that turning it off for small penetration depths d is helpful. If you wish, try other tweaks to bias (including just turning it off!).

This should be the scenario=5 that runs by default for the code you hand in (rigidbody2d_seq.py)

(b) scenario=6: Moving target. Repeat scenario 5 but have everything moving to the left (x-direction with a speed of -1.0).

This is a good test of the physical principle of Galilean invariance. Galileo posited that physics was the same regardless of whether things were moving. The trick is that *everything* should be moving (including the air to avoid creating drag).   The extended version of this is relativistic frame invariance. To an observer in that frame, only dealing with physics operating in that frame, physics is the same regardless of the relative speed of that frame to any other. The typical thought experiment is when you are on a perfectly smooth plane going straight, you can't tell that you are going at 900 km/h.

Passing this test means your physics engine can deal with a character on a moving platform or moving vehicle or similar cases.   You should effectively get the same results as for 5 (a) – a stable stack.